EfficientDet-D0 with EA-StrongSORT for Multi-Perspective Surgical Tool Tracking on CholecTrack20

Abdullah and Shamsa

May 2026

Abstract

This report presents an EfficientDet-D0 and EA-StrongSORT pipeline for multi-perspective surgical tool tracking on CholecTrack20. The study first documents a YOLO11 baseline, then replaces the detector with EfficientDet-D0, an EfficientNet-B0 plus BiFPN detector, and integrates it with EA-StrongSORT for tracking across visibility, intracorporeal, and intraoperative trajectory definitions. The selected EfficientDet-D0 detector achieved AP0.5 = 44.4, AP0.75 = 31.2, and AP0.5:0.95 = 27.9. The full tracking run produced MOTA values of 20.974, 19.641, and 19.314 for visibility, intracorporeal, and intraoperative perspectives respectively. The contribution is a reproducible EfficientNet-family detector-to-tracker baseline with complete metric logging and analysis, rather than a claim of state-of-the-art performance.

1. Introduction

Surgical tool tracking in laparoscopic video is a difficult multi-object tracking problem because instruments frequently leave the camera view, are partially occluded, overlap, and appear under smoke, blur, blood, reflection, and lens fouling. CholecTrack20 addresses this by providing multi-perspective annotations for surgical instruments, including visibility, intracorporeal, and intraoperative track identities. The goal of this project was to extend the CholecTrack20 research direction with an EfficientNet-family detection front end followed by EA-StrongSORT tracking.

2. Contribution

The implemented contribution has four parts. First, the project establishes a documented YOLO11 baseline. Second, it introduces EfficientDet-D0 as the active detector, selected because it is an EfficientNet-family object detector and is feasible on an RTX 2060 GPU. Third, it connects EfficientDet-D0 detections directly to EA-StrongSORT without relying on BoxMOT detector-name routing. Fourth, it adds tracking evaluation against CholecTrack20 JSON track IDs and exports reproducible CSV and Markdown logs for detector and tracker experiments.

3. Dataset and Evaluation Context

CholecTrack20 contains 20 laparoscopic cholecystectomy videos split into 10 training, 2 validation, and 8 testing videos. The dataset provides seven tool classes: grasper, bipolar, hook, scissors, clipper, irrigator, and specimen bag. For tracking, it provides three identity definitions: visibility trajectory, intracorporeal trajectory, and intraoperative trajectory. The public repository reports detector AP metrics and tracking metrics including MOTA, MOTP, IDF1, and HOTA-style measures. Our report matches the public detector and tracker columns where possible, while noting that the tracking evaluator implemented here is an internal class-aware IoU evaluator and not an official TrackEval leaderboard submission.

4. YOLO11 Baseline and Switch Rationale

The project began with YOLO11 because it is a modern one-stage detector and gave a fast baseline for the detection-to-tracking pipeline. Several YOLO11 configurations were tested, including SGD at image sizes 640 and 768, AdamW at image size 640, and quick partial-dataset runs for pipeline validation. The best YOLO11 strict localization run was final_yolo11_img768_sgd, with mAP50-95 = 0.2670 in the training logs and paper-style AP0.5:0.95 = 26.6. Although increasing image size from 640 to 768 gave a small improvement, the gain was not large enough to justify continuing only with YOLO11. The doctor/supervisor feedback also requested an EfficientNet-family alternative. Since plain EfficientNet is a classifier, the project selected EfficientDet-D0, which uses an EfficientNet-B0 backbone and is a real object detector. Therefore, YOLO11 was kept as the documented baseline, and EfficientDet-D0 became the active detector for the final EA-StrongSORT contribution.

YOLO11 hyperparameter experiments

Run	Key settings	Best epoch	Precision	Recall	mAP50	mAP50-95	Decision
final_yolo11_img768_sgd	SGD, imgsz=768, batch=8, lr0=0.01, patience=30	42	0.5595	0.4683	0.4077	0.2670	Best strict localization YOLO11 run
final_yolo11_patience30	SGD, imgsz=640, batch=16, lr0=0.01, patience=30	31	0.5700	0.4715	0.4108	0.2631	Best mAP50/balanced YOLO11 run
final_yolo11_adamw_lr001	AdamW, imgsz=640, batch=16, lr0=0.001, patience=40	44	0.5573	0.4716	0.4035	0.2519	Worse than SGD
sweep_20260503_sgd_lr001_img640	SGD, imgsz=640, batch=16, lr0=0.01	31	0.5638	0.4521	0.4048	0.2627	Close to main SGD baseline
quick_yolo_test	SGD, imgsz=384, batch=16, epochs=10, fraction=0.35	10	0.3910	0.3338	0.2867	0.1608	Pipeline test only

Best YOLO11 paper-style detector row

Detection model	AP0.5	AP0.75	AP0.5:0.95	Grasper	Bipolar	Hook	Scissors	Clipper	Irrigator	Bag	FPS
YOLO11-img768-SGD	40.3	29.6	26.6	42.1	45.8	50.0	23.6	59.3	3.4	57.8	62.9

Why the detector was switched to EfficientDet-D0

Observation	Evidence	Consequence
YOLO11 improvements saturated	imgsz 640 to 768 improved mAP50-95 only from 0.2631 to 0.2670	More YOLO-only tuning had diminishing returns
AdamW did not improve YOLO11	final_yolo11_adamw_lr001 reached mAP50-95 0.2519	SGD remained the better YOLO11 optimizer
Strict localization remained weak	Best YOLO11 paper-style AP0.5:0.95 was 26.6	Detector quality was unlikely to support strong tracking
Irrigator remained weak	YOLO11 irrigator AP0.5 was 3.4	Class imbalance/appearance issues required a different detector experiment
Supervisor direction favored EfficientNet-family study	EfficientNet was requested as an alternative contribution	EfficientDet-D0 was chosen as an EfficientNet-family object detector

5. Method

The final detector is EfficientDet-D0 using a COCO-pretrained EfficientNet-B0 backbone and BiFPN detection head. The selected training run used image size 640, AdamW optimization, learning rate 0.0002, weight decay 0.0001, batch size 6, gradient accumulation of 2, six workers, 150 total epochs, and validation every three epochs. The best checkpoint occurred at epoch 63, after which validation performance declined slightly. For tracking, EfficientDet detections were converted to [x1, y1, x2, y2, confidence, class] format and passed into StrongSORT with OSNet ReID weights. Predictions were exported in MOT text format and evaluated against the three CholecTrack20 perspective-specific track ID fields.

6. Detector Experiments

Detector tuning used short screening runs before longer training. The 150-epoch EfficientDet-D0 run at image size 640 and learning rate 0.0002 produced the best AP0.5:0.95 and was selected for tracking. It improved over our YOLO11 baseline in AP0.5, AP0.75, and AP0.5:0.95, but it remained below the strongest public CholecTrack20 YOLOv7-v10 detector benchmarks.

Detector hyperparameter screening

Selected	Run	Total epochs	Best epoch	AP0.5	AP0.75	AP0.5:0.95	FPS
yes	efficientdet_d0_640_adamw_lr2e4_long	150	63	44.4	31.3	27.9	20.9
no	efficientdet_d0_512_adamw_lr2e4_batch12_workers6_eval3	100	96	46.6	28.5	27.2	28.8
no	screen_effdet_d0_640_lr2e4	30	30	42.9	29.3	26.6	
no	screen_effdet_d0_640_lr1e4	30	30	41.2	28.9	25.7	
no	screen_effdet_d0_lr1e4	30	25	41.7	26.3	24.8	
no	screen_effdet_d0_sgd_lr1e3	30	20	37.3	22.9	21.8	

Detector comparison with CholecTrack20-style AP columns

Detector	AP0.5	AP0.75	AP0.5:0.95	FPS	Source
YOLOv7	80.6	62.0	56.1	20.6	CholecTrack20 GitHub
YOLOv8	79.1	62.4	55.6	29.0	CholecTrack20 GitHub
YOLOv9	80.2	62.6	56.5	23.7	CholecTrack20 GitHub
YOLOv10	80.1	62.1	55.8	28.6	CholecTrack20 GitHub
FCOS	43.5	31.5	28.1	7.7	CholecTrack20 GitHub
YOLO11-img768-SGD	40.3	29.6	26.6	62.9	Our baseline

Detector	AP0.5	AP0.75	AP0.5:0.95	FPS	Source
EfficientDet-D0-640-AdamW-lr2e4	44.4	31.2	27.9	20.9	Our contribution

Per-class detector comparison

Model	Grasper	Bipolar	Hook	Scissors	Clipper	Irrigator	Bag
YOLO11-img768-SGD	42.1	45.8	50.0	23.6	59.3	3.4	57.8
EfficientDet-D0-640-AdamW-lr2e4	47.3	50.8	60.3	33.4	60.5	0.6	57.6

Visual challenge AP comparison

Model	Bleeding	Blur	Smoke	Crowded	Occluded	Foul Lens	Trocar
YOLO11-img768-SGD	15.1	100.0	43.5	22.8	46.5	12.7	22.5
EfficientDet-D0-640-AdamW-lr2e4	19.2	75.2	46.9	24.4	52.0	17.1	47.3

7. Tracking Experiments

The selected EfficientDet-D0 checkpoint was evaluated with EA-StrongSORT on all eight CholecTrack20 test videos. The benchmark produced 768,555 MOT-format prediction rows across the test set. Tracking evaluation used annotated CholecTrack20 frames only with class-aware IoU >= 0.5 matching. MOTP remained high at 82.536, showing that matched boxes were localized reasonably well, but MOTA and IDF1 were limited by missed detections, false positives, and identity fragmentation.

Implemented EA-StrongSORT tracking metrics

Perspective	MOTA	IDF1	MOTP	Precision	Recall	IDSW	FP	FN
Visibility	20.974	24.172	82.536	68.692	58.952	3332	8059	12312
Intracorporeal	19.641	12.158	82.536	68.692	58.952	3732	8059	12312
Intraoperative	19.314	6.298	82.536	68.692	58.952	3830	8059	12312

8. Comparison with the CholecTrack20 GitHub Benchmark

Against the public detection leaderboard, EfficientDet-D0 is comparable to older detector families such as FCOS on AP0.5:0.95 but far behind the strongest YOLOv7-v10 benchmarks. Against the public tracking leaderboard, our EA-StrongSORT pipeline does not outperform Bot-SORT or ByteTrack. However, it provides a new EfficientNet-family detector-to-tracker baseline and shows that detector quality is the main bottleneck before tracker parameter optimization can be expected to help substantially.

Comparison with selected CholecTrack20 GitHub tracking entries

Perspective	Model	MOTA	IDF1	MOTP	FPS	Source
Visibility	Bot-SORT	72.0	41.4	83.7	8.7	CholecTrack20 GitHub
Visibility	ByteTrack	69.3	36.8	84.0	16.4	CholecTrack20 GitHub
Visibility	SORT	21.4	13.4	83.3	19.5	CholecTrack20 GitHub
Visibility	EfficientDet-D0 -> EA-StrongSORT	21.0	24.2	82.5	-	Our contribution
Intracorporeal	Bot-SORT	70.0	18.9	83.7	8.7	CholecTrack20 GitHub
Intracorporeal	ByteTrack	67.4	16.9	84.0	16.4	CholecTrack20 GitHub
Intracorporeal	EfficientDet-D0 -> EA-StrongSORT	19.6	12.2	82.5	-	Our contribution
Intraoperative	Bot-SORT	69.6	10.2	83.7	8.7	CholecTrack20 GitHub
Intraoperative	ByteTrack	67.0	9.5	84.0	16.4	CholecTrack20 GitHub
Intraoperative	EfficientDet-D0 -> EA-StrongSORT	19.3	6.3	82.5	-	Our contribution

9. Limitations

The main limitation is detector quality. The selected EfficientDet-D0 model reaches AP0.5:0.95 = 27.9, whereas the strongest public CholecTrack20 detectors exceed 55 on the same AP column. The irrigator class remains especially weak, suggesting class imbalance and visual ambiguity. The tracker also produces many identity switches, especially under intracorporeal and intraoperative identity definitions, where out-of-view and out-of-body identity continuity is stricter than frame-visible tracking.

10. Future Work

Future work should prioritize improving detection through stronger EfficientDet variants if GPU memory allows, class rebalancing, hard-example mining, and additional data augmentation for rare tools. After detector quality improves, EA-StrongSORT can be tuned through confidence thresholds, max age, ReID distance thresholds, and motion compensation behavior. A final official comparison should use the original CholecTrack20 TrackEval adaptation so that HOTA and leaderboard values are directly comparable.

11. Conclusion

The project successfully implements and documents an EfficientDet-D0 to EA-StrongSORT pipeline for CholecTrack20. While it does not exceed the strongest public benchmark results, it adds a reproducible EfficientNet-family baseline, demonstrates end-to-end multi-perspective tracking, and identifies detector quality and identity fragmentation as the key barriers to stronger performance.

References

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